

BHOJPURI ASR: 3 LOCAL MODELS & DATASET SPECIFICATION REPORT

Detailed Checkpoint Progression, Dataset Rows, Column Features & Word Error Rate (WER) Analysis

Table 1: Model 1 — Whisper-Small (Full Fine-Tuned)

Dataset Rows / Hours: 103,746 Training Rows (~150.2 Hours) | Eval: 1,054 Rows (~4.8 GB Arrow cache)

Dataset Columns (6 Features): audio (16kHz waveform array), file (wav path), transcript (Devanagari text), duration (sec), speaker_id, topic_domain

Architecture & Params: Seq2Seq Transformer (242.6M Total, **242.6M / 100% Trainable**) | Saved Path: models/bhojpuri-whisper-small-full

Checkpoint (Step)	Epoch	Eval Loss	WER (%) ↓	Status & Dataset Progression Notes
Step 100	0.010	0.6940	71.50%	Initial adaptation starting point on 103,746 rows
Step 900	0.086	0.3724	53.00%	Rapid early phonetic learning across Devanagari tokens
Step 1,700	0.162	0.3420	49.19%	Broke sub-50% WER threshold
Step 3,000	0.286	0.3301	49.46%	Stable convergence across multi-speaker acoustic rows
Step 5,200	0.496	0.2985	45.28%	Significant vocabulary alignment on 6 topic domains
Step 7,700	0.735	0.2813	43.48%	Steady loss reduction on telephony & studio speech
Step 8,700	0.830	0.2776	42.74%	Improved complex phrasing and compound word accuracy
Step 10,500	1.002	0.2668	41.66%	Passed all 103,746 training rows (1.0 full epoch)
Step 10,800	1.030	0.2712	41.56%	Checkpoint-10800 saved on local disk
Step 13,600	1.298	0.2633	41.65%	Minor loss fluctuations near global minimum
Step 14,500	1.383	0.2604	41.06%	Approaching optimal cross-entropy loss
■ Step 14,900	1.422	0.2655	40.58%	All-Time Best Full Fine-Tuning Checkpoint (Saved on disk)

Table 2: Model 2 — Whisper-Small + LoRA (Parameter-Efficient PEFT)

Dataset Rows / Hours: Trained on the same 103,746 Rows (~150h) starting from base Checkpoint-14900 | Eval: 1,054 Rows

Dataset Columns / Input Tensors: input_features (80-channel log-Mel spectrogram), labels (BPE token IDs)

LoRA Config & Trainable Params: Rank $r=8$, $\alpha=16$, modules: q_proj, v_proj | **884,736 / 242.6M (0.36%) Trainable**

Saved Path: models/LORAmoel/lora-merged-final

Checkpoint (Step)	Epoch	Eval Loss	WER (%) ↓	Status & LoRA Progression Notes
Step 100	0.019	0.2591	40.59%	Initial LoRA baseline (inherited from Checkpoint-14900)
Step 300	0.057	0.2546	39.30%	Rapid early adapter improvement across attention matrices
Step 700	0.134	0.2530	38.99%	First time breaking sub-39% WER barrier in project
Step 800	0.153	0.2521	39.33%	Stable validation checkpoint
Step 1,000	0.191	0.2514	39.81%	Temporary greedy decoding variance
Step 1,400	0.267	0.2502	38.89%	Approaching global peak accuracy
■ Step 1,800	0.343	0.2465	38.86%	All-Time Lowest WER across entire Bhojpuri project (Best Step)
Step 2,100	0.401	0.2463	39.39%	Cosine learning rate scheduler entered decay phase
Step 2,700	0.515	0.2465	39.13%	Sustained high accuracy plateau on test split
Step 3,500	0.668	0.2471	40.17%	Completed 3,500 steps (lr decayed to 5.88e-08)
■ Final Merged	—	0.2465	38.91%	Final standalone fused model (+1.67% improvement over full FT)

Table 3: Model 3 — Vakyansh Wav2Vec 2.0 (bhom_60 CTC)

Dataset Rows / Hours: Pre-trained on ~36,000 Rows (~60.0 Hours) Bhojpuri speech (ULCA / IISc RESPINS & SYSPIN)

Dataset Columns (3 Features): audio_filepath (16kHz 1D waveform), duration (sec), text (65 Devanagari character tokens)

Architecture & Params: Wav2Vec 2.0 Base + CTC Head (~95M Total, ~95M Trainable) | Saved Path: models/vakyansh-bhojpuri

Benchmark Test Split	Samples / Rows	WER (%) ↓	Inference Speed	Acoustic & Error Characteristics
Studio Speech Split (IISc SYSPIN)	610 Audio Rows (57.9%)	102.96%	~25 files/sec	Clean acoustic room; high character substitution & token insertion
Field Mobile Split (AI4Bharat)	444 Audio Rows (42.1%)	114.88%	~25 files/sec	High error rate on ambient village background & telephone bandwidth
Total Benchmark Eval Corpus	1,054 Audio Rows	108.22%	■ ~25 files/sec	Evaluated on data/merged_bhojpuri/eval (~5x faster CTC throughput)

Executive Summary & Dataset Insights

- Dataset Volume Impact:** Training on 103,746 rows (150h) enabled Whisper-Small to reduce WER from 75% down to 40.58%.
- LoRA Parameter & Sample Efficiency:** LoRA achieved the lowest project WER (38.91% final / 38.86% peak) updating only 884K parameters (0.36%) on the exact same 103,746 rows.
- CTC vs Autoregressive Trade-Off:** Vakyansh provides ~5x faster throughput (~25 audio files/sec) but suffers from high WER (108.22%) due to character insertion errors and lack of an autoregressive language model decoder.